

Linking EU ETS installations to ENTSOE-Power Plants and IEP Facilities

Jan Abrell^{*} Mirjam Kosch[†] Leonard Stimpfle[§]

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1 Introduction

Industrial activities in the European Union are covered by multiple complementary reporting obligations, most notably

- the EU Emissions Trading System (Directive 2003/87/EC, [2003](#), EU ETS),
- the ENTSOE Transparency Platform (Commission Regulation (EU) No 543/2013, [2013](#)), and
- the Industrial Emissions Portal (Regulation (EU) 2024/1244, [2024](#), IEP) combining the E-PRTR (Regulation (EC) No 166/2006, [2006](#)) and Industrial Emissions Directive (Directive 2010/75/EU, [2010](#)).

Although a single industrial plant often reports under multiple obligations, there are no common identifiers across the corresponding databases. This project addresses this shortcoming by providing links between the different databases.

The EU ETS regulates greenhouse gas emissions of certain activities under a classical cap-and-trade system. Annual verified emissions and allowance transactions under the EU ETS are provided in the European Transaction Log (EUTL) (Directorate-General for Climate Action, [2024](#)) which is the administrative backbone of the EU ETS. The

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[†]Chair of Energy Economics, University Basel, Faculty of Business and Economics, Peter-Merian-Weg 6, CH-4002 Basel, jan.abrell@unibas.ch

[‡]mirjam.kosch@outlook.com

[§]Ghent University, Faculty of Economics and Business Administration, Sint-Pietersplein 7, BE-9000 Ghent leonard.stimpfle@ugent.be

EUTL, however, only provides emission data and information on the transfer of emission allowances.

Linking the EUTL with other databases therefore presents significant opportunities to improve the data availability on industrial facilities in the Europe. For example, power producers are the largest emitters in the EU ETS and linking the EUTL to ENTSOE can provide granular data on their production decisions. Secondly, the EU ETS regulates and reports on specific greenhouse gases such as carbon dioxide or nitrous oxides. The Industrial Emissions Portal (IEP) complements the EUTL by reporting releases of a much broader set of pollutants to air, soil, and water, thereby providing a more nuanced picture of industrial activities in Europe.

The remainder of this note is structured as follows. Section 2 discusses the linking of the EUTL with ENTSOE and Section 3 the linking of the EUTL with the IEP.

2 Linking the EU ETS with ENTSOxE

We link installations regulated under the EU ETS to units that generate electricity as provided by the ENTSOE transparency platform (ENTSOE, 2024).¹ In the EUTL installations are identified by a unique installation identifier. In the ENTSOE database, power plants are identified on two different levels using Energy Identification Codes (EIC). The *generation unit* identifies the lower unit of generation and several generation units can form a *production unit*. There is, however, no one-to-one relation between installations and production or generation units. To resolve the possibility that several installations might relate to several generation or production units — to resolve the n:m mapping — we introduce artificial power plants and establish a unique link from EU ETS installations and ENTSOE generation units to these artificial power plants.

2.1 Linking procedure

The linking of the two datasets is performed manually. In a first step, we use the JRC power plant database (Hidalgo Gonzalez et al., 2019; Kanellopoulos, 2019) that provides a link from ENTSOE to several other database including the former version of IEP database. We use the IEP and the linking to the EUETS to establish a first set of links.

The remaining matches are established using a geolocation matching based on the latitude and longitude provided by Abrell (2024) for the EUETS installation and ENTSOE (2024) for the ENTSOE units. The geographical matching provides a first set of candidate links. The final selection is done using additional information from the following sources:

- Global Energy Monitor (2024) provides information on power plants, their individual units including historical changes.
- Beyond Fossil Fuels (2024) provides a list of coal power plants and their emissions.

¹The linking was produced in the time from 2023 to 2025. The latest EU ETS data batched used for linking was the data batch for 2024 as obtained from euets.info

- The power plant matching tool provides a matching from ENTSOE to several other databases (Gotzens et al., 2019, also see the [github repository](#))
- Open Power System Data (2020) provides a list of European conventional power plants partly including EIC identifiers and additional information on address and plant specifications.

2.2 EU ETS to ENTSOE Linking

The EU ETS to ENTSOE linking is provided in three files:

1. *plants_final.csv* introduces power plants. It provides the unique identifier of the plant together with the fuel type of the plant that is derived from the ENTSOE data. Possible fuel identifiers are *coal*, *lignite*, *gas*, *waste*, *oil*, *biomass*, and *coalgas*.²
2. *eutl_2_plants_final.csv* establishes the linking of the EU ETS installations to these plants.
3. *entsoe_2_plants_final.csv* links the ENTSOE generation units to the plants.

Table 2 provides a first overview of the linking. It provides the number of plants and associated installations and ENTSOE units by the fuel assigned to the plants. Overall there are 481 plants representing 594 EU ETS installations and 1011 (567) ENTSOE generation (production) units. The majority of these plants, roughly 50%, use natural gas as input fuel, followed by coal and lignite.

Table A.1 shows the number of plants by country and fuel used. Not surprisingly, Germany as largest electricity consumer shows the highest number of plants. As the example of Italy however shows, the number of plants per country is also influenced by the fuels used as gas plants usually have lower installed capacity.

3 Linking the EU ETS with the IEP

We link installations in the EU ETS Union Registry (Directorate-General for Climate Action, 2024) to facilities in the Industrial Emissions Portal (IEP) (European Environment Agency, 2024). Each EU ETS installation identifier is linked to exactly one IEP facility identifier. IEP facilities are labeled in accordance with the Infrastructure for Spatial Information in Europe (INSPIRE) regulation (Directive 2007/2/EC, 2024). This means that one EUTL installation is linked to exactly one IEP facility and that one IEP facility is linked to one or several EUTL installations.

3.1 Linking procedure

We use probabilistic record linkage (Enamorado et al., 2019; Fellegi & Sunter, 1969) as implemented in the Python library “Splink” (2025) (Linacre et al., 2022). The probabilistic linkage model predicts whether a given EUTL installation and IEP facility

² *coalgas* corresponds to the ENTSOE category of *coal derived gases*.

Table 1: Linking EU ETS and ENTSOE: Column Descriptions

Column	Description	Type
<i>plants_final.csv</i>		
<i>plant_id</i>	Unique identifier of the plant. It is derived as the 2 digit ISO code of the country in which the plant is located and the name of the plant (mostly inferred from the ENTSOE data) concatenated by an underscore.	string
<i>fuel</i>	An identifier for the fuel the plant uses.	string
<i>eutl_2_plants_final.csv</i>		
<i>installation_id</i>	Unique identifier of the EU ETS installation; constructed as the 2-digit ISO code for the registry (i.e., country) and the installation identifier in the EUTL (which is unique only within registry).	string
<i>plant_id</i>	Unique identifier of the plant.	string
<i>entsoe_2_plants_final.csv</i>		
<i>eic_g</i>	EIC code of the generation unit.	string
<i>name_g</i>	Name of the generation unit.	string
<i>eic_p</i>	EIC code of the production unit.	string
<i>name_p</i>	Name of the production unit.	string
<i>plant_id</i>	Unique identifier of the plant.	string

Note: The table describes the columns of the files that provide the linking of the EU ETS installations to the ENTSOE power plants. Since multiple installations might map to one ENTSOE unit and vice versa, the *plants_final.csv* introduces power plants. *eutl_2_plants_final.csv* establishes the linking of the EU ETS installations to these plants. Likewise, *entsoe_2_plants_final.csv* links the ENTSOE generation units to the plants.

represent in fact the same entity based on (fuzzy) comparisons of several matching variables available for both records such as installation name, operator name, geolocation (longitude/latitude), address, activity and industry.³ A key advantage of probabilistic record linkage is the direct quantification of the match uncertainty. For each EUTL installation we select the IEP facility with the highest match probability above a minimum match probability threshold of 50%.

³We apply various cleaning routines to the match variables before linking the datasets. This includes standard operations applied across all string fields such as transliteration to the Latin alphabet and removing accents and special characters as well as variable-dependent operations such as removing legal entity descriptors in company names (e.g., “Limited” or “GmbH”).

Table 2: Overview: Linking of EUETS and ENTSOE

Fuel	Plants	Installations	Generation Units	Production Units
gas	253	310	480	281
coal	124	161	269	156
lignite	42	54	158	55
oil	32	37	70	45
biomass	20	20	21	19
coalgas	9	11	12	10
waste	1	1	1	1
Total	481	594	1011	567

The table shows the number of linked entities by fuel. It shows the number of plant by fuel together with the associated EUETS installations and ENTSOE generation and production units.

3.2 EU ETS to IEP Linking

The linking between EU ETS installations and IED facilities is provided in a single file `eutl_2_iep_final.csv` the contents of which are described in Table 3

Table 3: Linking EU ETS and IED: Column Descriptions

Column	Description	Type
<code>installation_id</code>	Unique identifier of the EU ETS installation; constructed as the 2-digit ISO code for the registry (i.e., country) and the installation identifier in the EUTL (which is unique only by registry)	string
<code>facility_inspire_id</code>	Unique INSPIRE identifier of the IEP facility.	string
<code>match_probability</code>	Match probability of the established link. Varies between 0.5 and 1 (50 and 100%).	float

Note: The table describes the columns of the file that provide the linking of the EU ETS installations to the IED facilities.

3.3 Overview: Linking of EU ETS and IEP

Table 4 presents the distribution of match probabilities of the links between the EUTL and the IEP. Overall, we match 12,284 EU ETS installations to 9,585 IEP facilities. Match probabilities are generally high with the 25th percentile already at 98%. Table A.2 provides an overview of matched installations and facilities by country.

We extensively validate the linkage model. First, we identify deterministic links for

Table 4: Match Probabilities [%]

Mean	97.6	25% Quartile	99.7
Standard deviation	7.3	50% Quartile	99.9
Minimum	50	75% Quartile	100
Maximum	100	Observations	12,284

Note: Table shows the descriptive statistics of the matching probabilities for the established links between EU ETS installations and IEP facilities.

455 EUTL installations based on common identifiers in the EUTL and the IEP.⁴ Second, we obtain 699 links from the AIDRES database (Catholic University Leuven et al., 2023). Third, we clerically label links for 100 EUTL installations. Across all test cases, we find a true positive rate in excess of 95% where most incorrect links are false negatives driven by very poor data quality. The high quality of the linkage model is further evidenced by comparing the carbon dioxide emissions reporting of linked installations between the EUTL and IEP (see Figure 1).

⁴Specifically, in the EUTL some installations report IEP facility identifiers in the field `eptr_id`. In the IEP, the table `3f2.ETSIdentifiers` contains identifiers of some EUTL installations. In total, we find 455 valid links, i.e., pairs of installations and facilities where the identifiers agree in both databases.

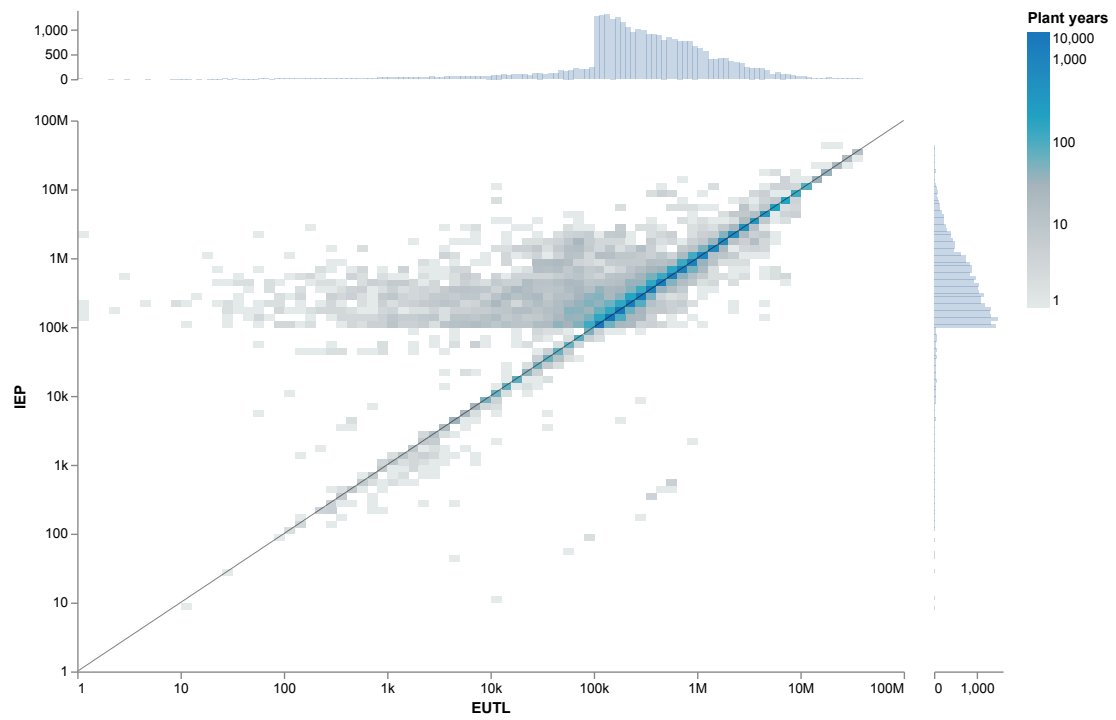


Figure 1: The figure presents a binned scatter plot of annual installation-level CO₂ emissions (in tCO₂) in the EUTL (x -axis) versus the IEP (y -axis). The number of installation-years in each bin is indicated by the color. The figure also presents the marginal emissions distributions for the EUTL (top histogram) and IEP (right histogram).

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4 Appendix: Matching overview by country

Table A.1: Number of power plants by country and fuel used

fuel	biomass	coal	coalgas	gas	lignite	oil	waste	Total
AT		3		8		1		12
BE	1	1	1	12				15
BG		1		1	4			6
CZ		1		1	7			9
DE		36	4	36	13	3	1	93
DK	3	8		2		1		14
EE						3		3
ES		15		27	1			43
FI	7	9		2		1		19
FR	1	9		16		7		33
GR				11	6	1		18
HU				5	1	3		9
IE	3	1		10		3		17
IT	1	9	3	71		5		89
LT				3				3
LU				1				1
LV				1				1
NL		7	1	21				29
NO				1				1
PL	1	19		7	4			31
PT		1		4				5
RO		2		8	4			14
SE	3	1		2		4		10
SI				1	1			2
SK		1		2	1			4
Total	20	124	9	253	42	32	1	481

Table A.2: Linked EU ETS installations and IED facilities by country

Country	Installations	Facilities
AT	222	172
BE	417	337
BG	131	120
CY	13	13
CZ	385	329
DE	2194	1456
DK	212	170
EE	46	37
ES	1175	1042
FI	533	262
FR	1179	1044
GB	1031	901
GR	138	129
HR	59	57
HU	206	170
IE	130	116
IS	7	7
IT	1226	1083
LI	1	1
LT	76	61
LU	22	18
LV	48	36
MT	6	3
NL	427	329
NO	127	114
PL	781	585
PT	271	226
RO	224	203
SE	803	388
SI	66	60
SK	128	116
Total	12284	9585